Database Management System (DBMS)

Previous Year Question Paper – 2020
Time: 3 Hours Maximum Marks: 100

Note: This is a sample/dummy placeholder question paper generated for demonstration purposes only. It does not represent an official examination paper. Some questions repeat across years with different wording, as is common in real previous-year question compilations.

Section A – Short Answer Questions (Answer any 5)

Q1. What is the difference between the WHERE clause and the HAVING clause in SQL? *(5 marks each)*

Q2. Define an Entity-Relationship diagram. What are entities, attributes and relationships? *(5 marks each)*

Q3. Define a database view. What are its advantages? *(5 marks each)*

Q4. What is a Database Management System? How is it different from a traditional file-based system? *(5 marks each)*

Q5. What do you understand by logical and physical data independence? Give examples. *(5 marks each)*

Section B – Medium Answer Questions (Answer any 4)

Q1. Describe the three levels of data abstraction (internal, conceptual, external) in DBMS. *(10 marks each)*

Q2. Discuss various types of keys (primary, candidate, foreign, composite, super) used in DBMS. *(10 marks each)*

Q3. Draw and explain an ER diagram for a Library Management System, identifying entities and relationships. *(10 marks each)*

Q4. What is a functional dependency? Explain its types with examples. *(10 marks each)*

Section C – Long Answer Questions (Answer any 3)

Q1. Model an Online Shopping System with an ER diagram, including Customer, Product, Order and Payment entities, and derive the relational schema. *(15 marks each)*

Q2. Take an unnormalized relation and normalize it step by step up to BCNF, explaining each step. *(15 marks each)*

Q3. Describe the phases of query processing in DBMS, including parsing, optimization and execution. *(15 marks each)*

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